# Day 03 Report: Digit Recognizer

Oct 30, 2026 · @Hiru

## Summary

The CNN recognised handwritten digits best, with 99.07% accuracy on 10,000 test images (about 93 mistakes).

- The MLP reached 97.59% (about 241 mistakes) and the SVM 95.94% (about 406 mistakes).
- The CNN wins because it reads the image as a picture and finds edges and shapes; the other two see only a flat list of pixels.
- With a GPU, the CNN trained in 35 seconds.

## Data and preparation

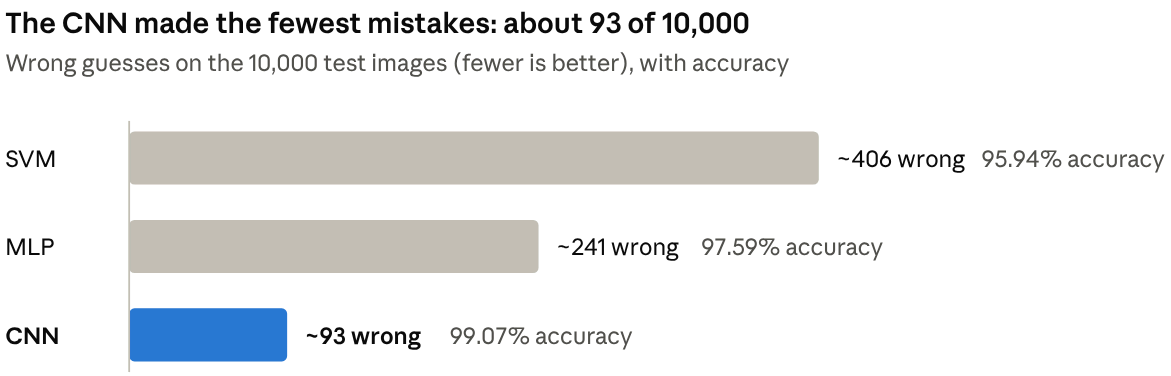
MNIST has 70,000 handwritten digits (0 to 9), each a 28 x 28 grayscale picture: 60,000 for training and 10,000 for testing. Each digit has about 6,000 training pictures, so the data is balanced.

| Step | What was done | Why |
|---|---|---|
| Scaling | Pixels divided by 255, so 0 to 255 becomes 0 to 1 | Small numbers help models learn faster and more steadily |
| Flat version | Each picture turned into one list of 784 numbers | SVM and MLP need a list |
| Picture version | Kept as 28 x 28 x 1 | The CNN needs the real picture shape |

# The three models

| Model | In simple words | Setup |
|---|---|---|
| SVM | Builds a curvy fence between groups of digits | RBF kernel, trained on 10,000 images because it is slow on all 60,000 |
| MLP | A simple brain that reads the picture as a long list | 784 inputs, 128 and 64 neurons, 10 outputs; 10 epochs |
| CNN | Slides small windows over the picture to find lines, then shapes | Two convolution layers (32 and 64 filters) with pooling, then dense layers; 10 epochs |

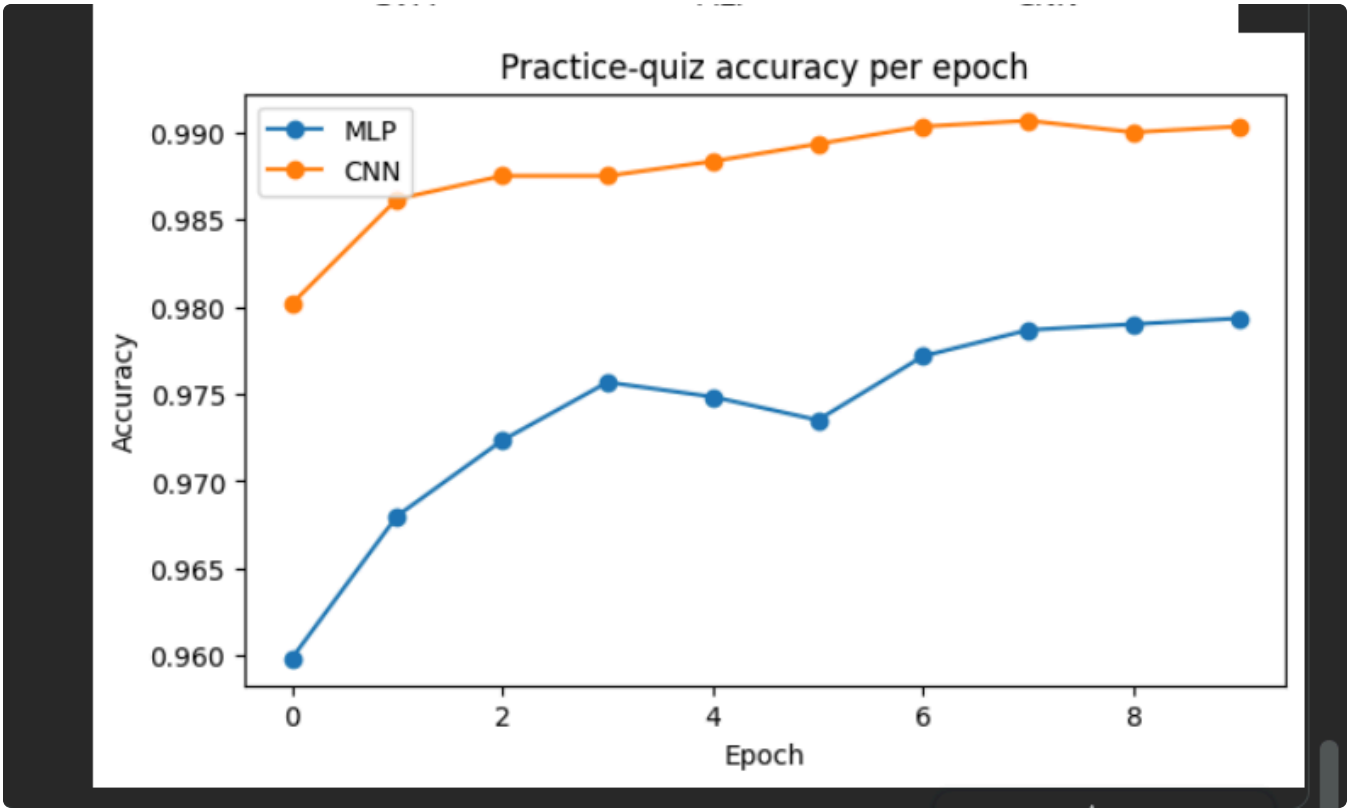
# Results




digit_recognizer.ipynb · MNIST test set, 10,000 images

Accuracy looks close (95.94% vs 99.07%), but counting mistakes shows the real gap: the CNN gets about 4 times fewer digits wrong than the SVM.

## Learning curves




The CNN (orange) was already at 98% after the first epoch and settled around 99%. The MLP (blue) started at 96% and reached about 98%. The CNN learned faster and finished higher.

## Conclusion and next steps

For images, the CNN is the right choice: it makes about 2.6 times fewer mistakes than the MLP and 4 times fewer than the SVM, and trains quickly on a GPU.

- Add data augmentation (small rotations and shifts) to make the CNN more robust.
- Add dropout to reduce overfitting.
- Look at the digits the CNN still gets wrong to see what confuses it.